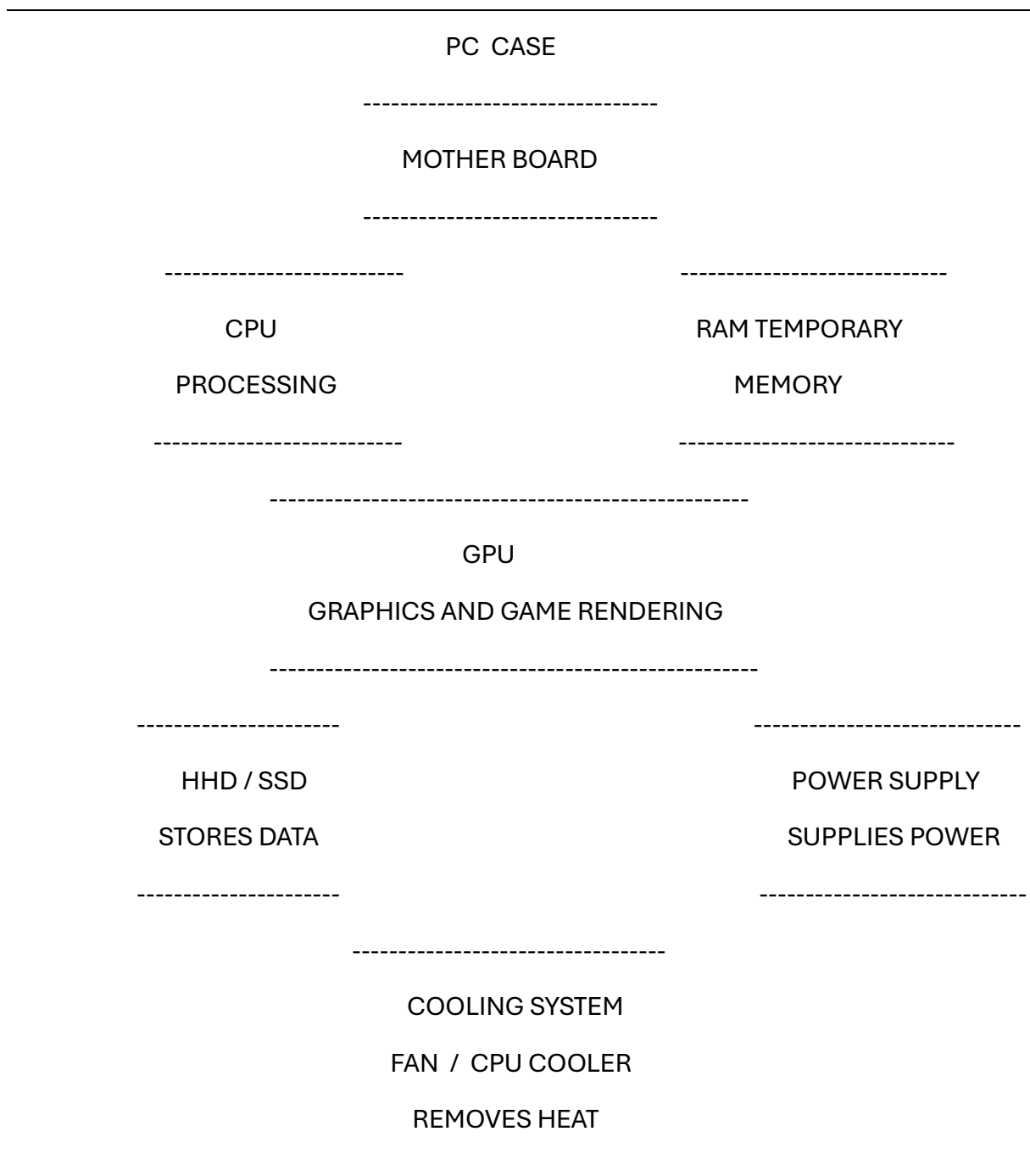


Overview of Hardware Components

1 - Draw a labeled diagram of a desktop PC showing the motherboard, CPU, RAM, GPU, storage (HDD/SSD), power supply, and cooling system. For each component, write its main function next to it.

- DESKTOP PC – INTERNAL COMPONENTS.



Component	Main Function
Motherboard	Connects and allows communication between all major components.
CPU	Processes instructions and performs calculations.
RAM	Temporarily stores data and programs currently being used.
GPU	Processes graphics and renders images, videos, and games.
HDD/SSD	Permanently stores the operating system, applications, and files.
Power Supply (PSU)	Converts and supplies electrical power to PC components.
Cooling System	Removes heat and keeps components at safe temperatures.

2 - Create a table listing each major PC component (motherboard, CPU, RAM, GPU, storage, power supply, cooling system), its typical location inside the case, and one example of how it connects or interacts with another component.

Hint: For example, "CPU sits in the motherboard socket and communicates with RAM via the memory slots."

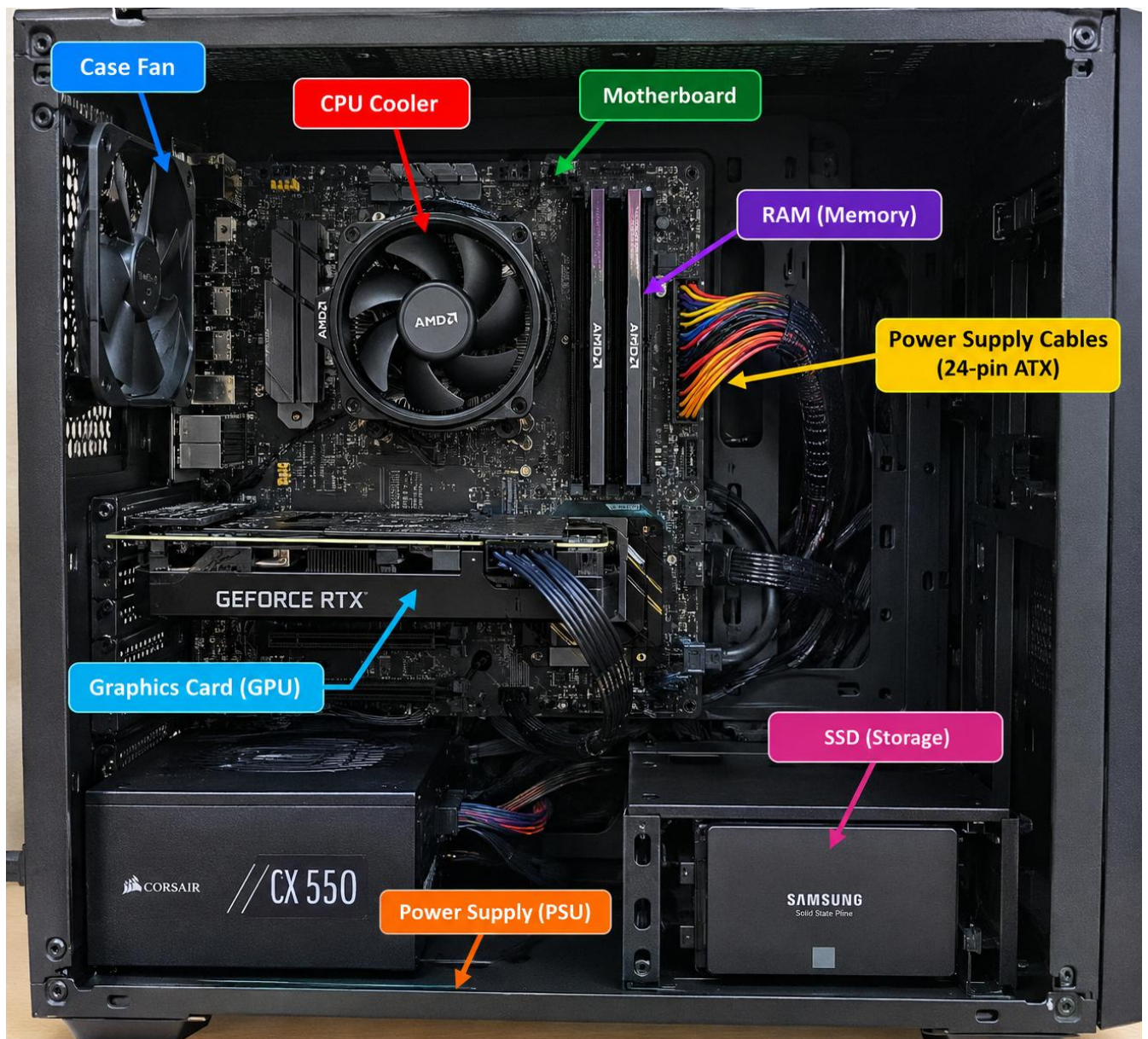
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Component	Typical Location Inside Case	Example of Connection/Interaction
Motherboard	Mounted vertically inside the case	Connects CPU, RAM, GPU, storage and other components.
CPU	In the CPU socket on the motherboard	CPU communicates with RAM through the motherboard's memory system.
RAM	RAM slots next to the CPU	Provides the CPU with fast temporary data storage.
GPU	PCIe slot on the motherboard	Receives data from the CPU/RAM and renders graphics.

Component	Typical Location Inside Case	Example of Connection/Interaction
Storage (HDD/SSD)	Drive bay or M.2 slot	Stores the operating system and sends game data to RAM when needed.
Power Supply	Usually at the bottom or top rear of the case	Supplies electrical power to the motherboard, GPU, drives, and other components.
Cooling System	CPU area and case fan positions	CPU cooler removes heat from the CPU; case fans move hot air out.

3 - Take a clear photo or screenshot of the inside of any desktop PC (your own, a friend's, or an image from the web) and use any image editor to digitally label at least 6 major components as discussed in class.

-> For this part, take a clear photo/screenshot of the inside of a desktop PC and use an editor such as Paint, PowerPoint, Canva, or any image-editing application.



Label at least these 6 components:

1. **Motherboard**
2. **CPU + CPU Cooler**
3. **RAM**
4. **GPU**
5. **HDD/SSD**
6. **Power Supply (PSU)**

You can also label the **case fans/cooling system** as a seventh component.

Suggested labels: Use arrows pointing from each component to a text box containing its name.

4 - Write a short explanation (3-4 sentences) describing the data flow when you open a game on your PC, mentioning the roles of CPU, RAM, storage, GPU, and motherboard.

Hint: Start from storage loading data, then CPU processing, RAM usage, and GPU rendering.

-> When you open a game, the **storage (HDD/SSD)** loads the game's files and sends the required data to **RAM**. The **CPU** processes the game instructions and manages game logic using the data available in RAM. The **GPU** receives graphics-related data and renders the game's images, animations, and 3D environment. The **motherboard** provides the connections that allow the CPU, RAM, GPU, storage, and other components to communicate with each other.